

EE6320 RF Integrated Circuits

Project: Mixer Design

Mixer Performance Summary Table

	Design Metric	Performance	Specification
Conversion Gain	Gain in the specified band [$f_{RF} = f_{LO}$]	20.134 dB	>20 dB
	Gain in the specified band [$f_{RF} = f_{LO} + 10MHz$]	20.132 dB	>20 dB
	Peak Gain flatness in specified band [Max-Min Gain]	0.002 dB	-
	3dB RF Bandwidth [From the plot of $f_{RF} = f_{LO}$]	5.762 GHz	-
Noise Figure	Integrated SSB Noise Figure for $f_{LO} = 3.5 GHz$	8.68 dB	≤ 12 dB
	RF Frequency Band over which SSB NF<10dB $f_{LO} = 3.5GHz$	3.5 GHz $\pm$ 553.23 KHz	
Linearity - IIP_2	Input power used for extrapolation	-40 dBm	-
	Power of Fundamental Tone at output (at chosen input power)	-19.86 dBm	-
	Power of IM_2 Tone at output (at chosen input power)	-97.38 dBm	-
	Extrapolated IIP_2	37.51 dBm	$\geq +20dBm$
Linearity - IIP_3	Input power used for extrapolation	-40 dBm	-
	Power of Fundamental Tone at output (at chosen input power)	-19.86 dBm	-
	Power of IM_3 Tone at output (at chosen input power)	-90.25 dBm	-
	Extrapolated IIP_3	-4.80 dBm	$\geq -5 dBm$
Power	Mixer DC power consumption [Excluding Bias]	1.29 mW	Minimize
	Bias circuit power consumption	4.68 μ W	Minimize

Other	Sum of all resistances [excluding bias]	0	-
	Sum of biasing resistances	200 kΩ	-
	Sum of all capacitances [Including AC coupling]	2 μF	-
	Sum of all inductances	0	-
	Load chosen (each R_load)	1120 Ω	-

	Differential Mixer Input Capacitance (C_gs Caps)	193.58 fF	-
	Simulator Used	Spectre	-

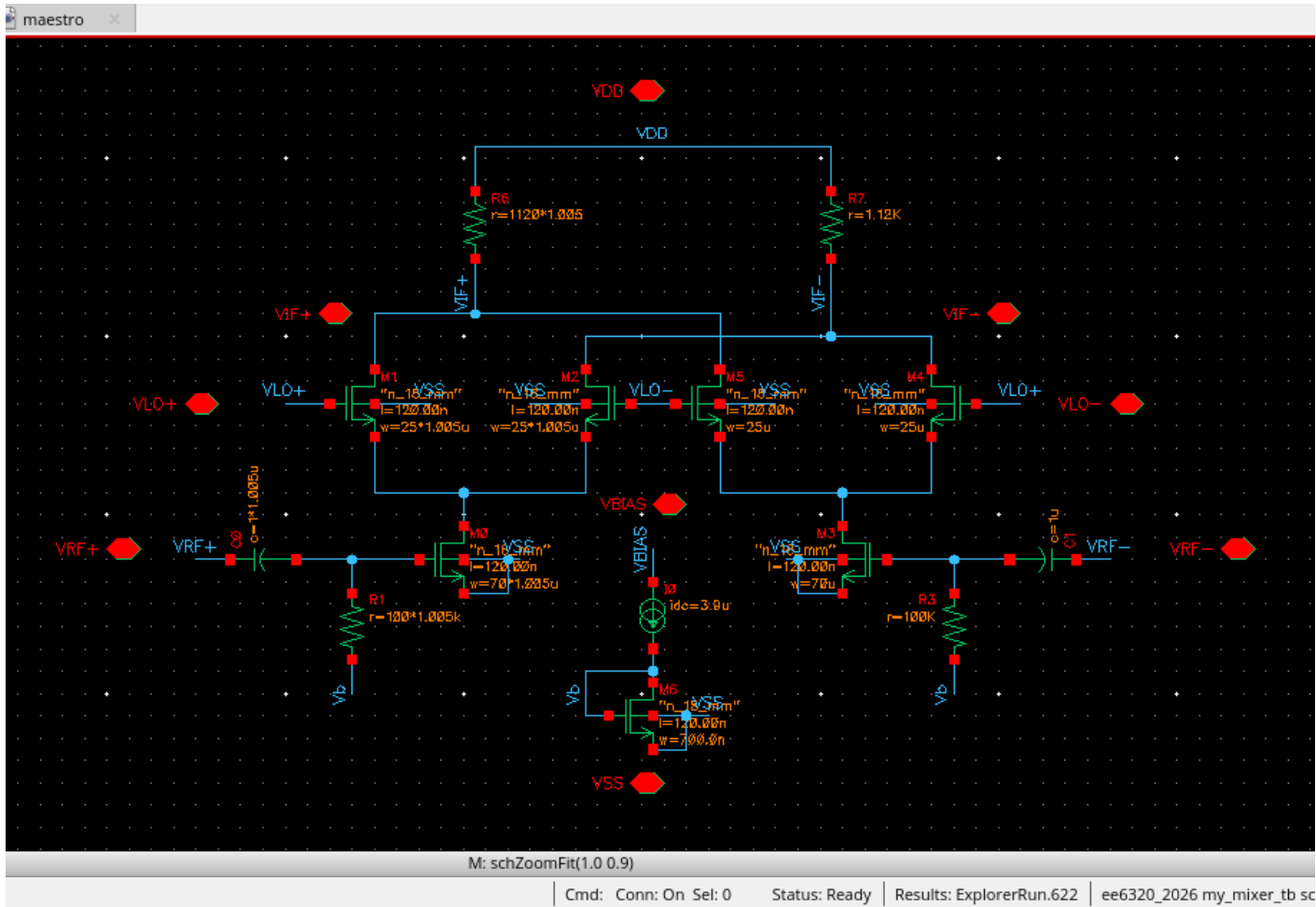
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LNA + Mixer Performance Summary Table

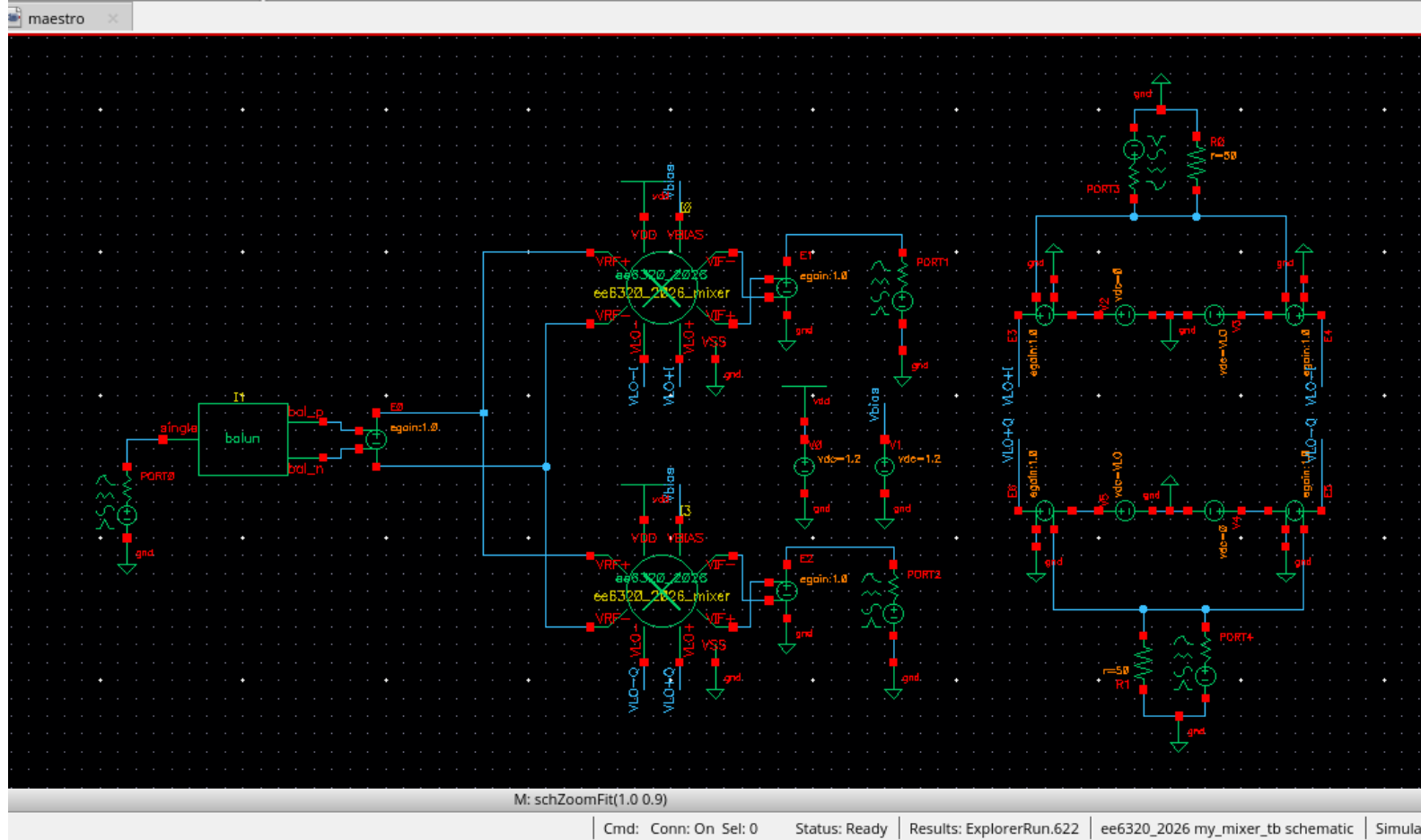
	Design Metric	LNA	Mixer	Cascade	
				Expected	Simulated
Conversion Gain	$f_{IN} = f_{LO}, f_{LO} = 3.5\text{ GHz}$	21.818 dB	20.134 dB	41.952 dB	33.746 dB
	$f_{IN} = f_{LO} + 10\text{MHz}, f_{LO} = 3.5\text{ GHz}$	21.825 dB	20.132 dB	41.957 dB	33.754 dB
	$f_{IN} = f_{LO} - 10\text{MHz}, f_{LO} = 3.5\text{ GHz}$	21.795 dB	20.137 dB	41.932 dB	33.721 dB
Noise Figure	$f_{IN} = f_{LO}, f_{LO} = 3.5\text{ GHz}$	1.806 dB	8.68 dB	3.08 dB	2.527 dB
	$f_{IN} = f_{LO} + 10\text{MHz}, f_{LO} = 3.5\text{ GHz}$	1.803 dB	8.68 dB	3.08 dB	2.527 dB
	$f_{IN} = f_{LO} - 10\text{MHz}, f_{LO} = 3.5\text{ GHz}$	1.809 dB	8.68 dB	3.08 dB	2.527 dB

Linearity IIP3	Input power used for extrapolation	-50 dBm	-50dBm	-	-50 dBm
	Power of Fundamental Tone at output (at chosen input power)	-28.185 dBm	-29.87 dBm	-	-16.197 dBm
	Power of IM_3 Tone at output (at chosen input power)	-108.647 dBm	-120.24 dBm	-	-71.501 dBm
	Extrapolated IIP_3	-9.77 dBm	-4.812 dBm	-26.7 dBm	-22.348 dBm
Power	Total power consumption [Excluding Bias]	8.7996 mW	1.29 mW	10.089 mW	10.093 mW
	Bias circuit power consumption	5.16 mW	4.68 μ W	5.164 mW	5.164 mW

Mixer Schematic



Mixer Testbench



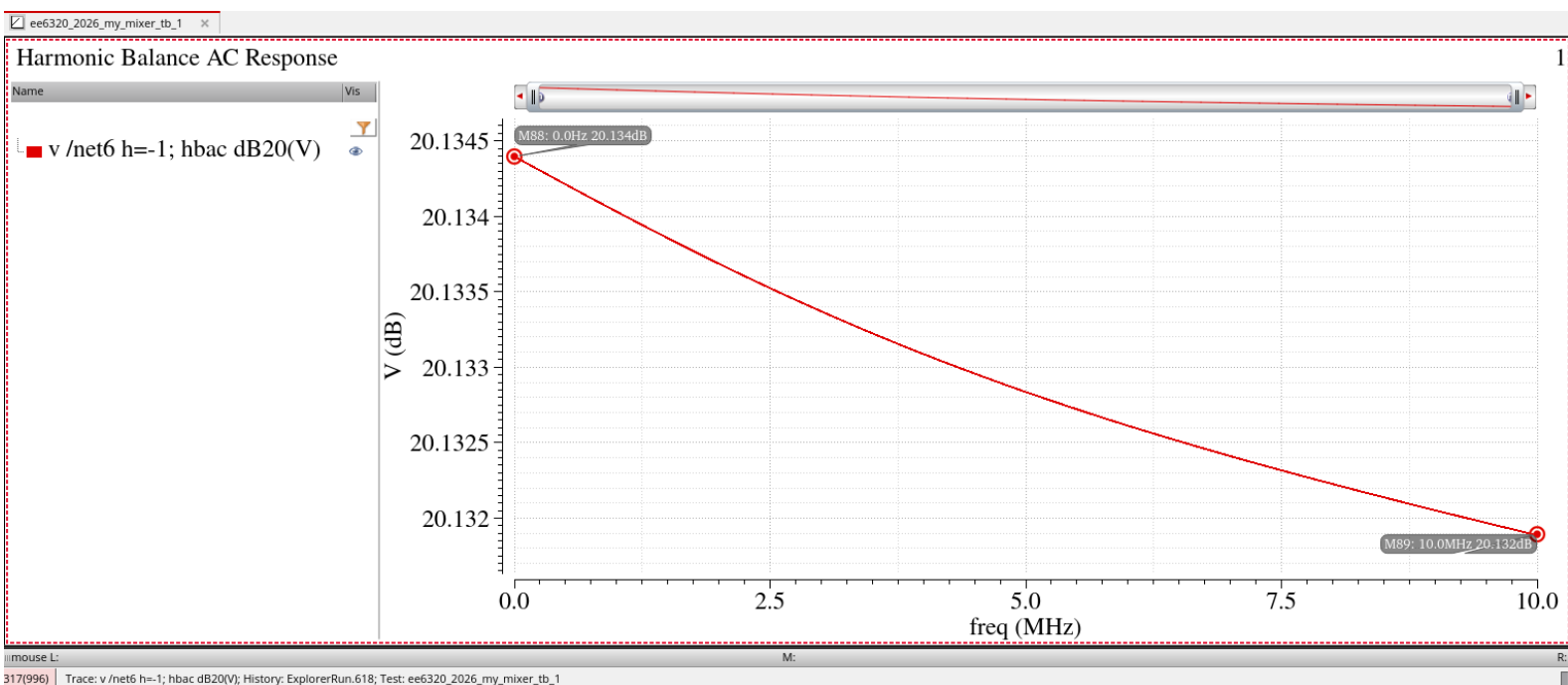
Design Variable Values

Design Variable	Value
Resistance (load, each side)	1120 Ω
Length of all MOS	0.12 μm
Width of Switch MOS (all 4)	25 μm
Width of Transconductance MOS (both MOS)	70 μm
Bias Current	3.9 μA
V_LO Amplitude	0.71 V

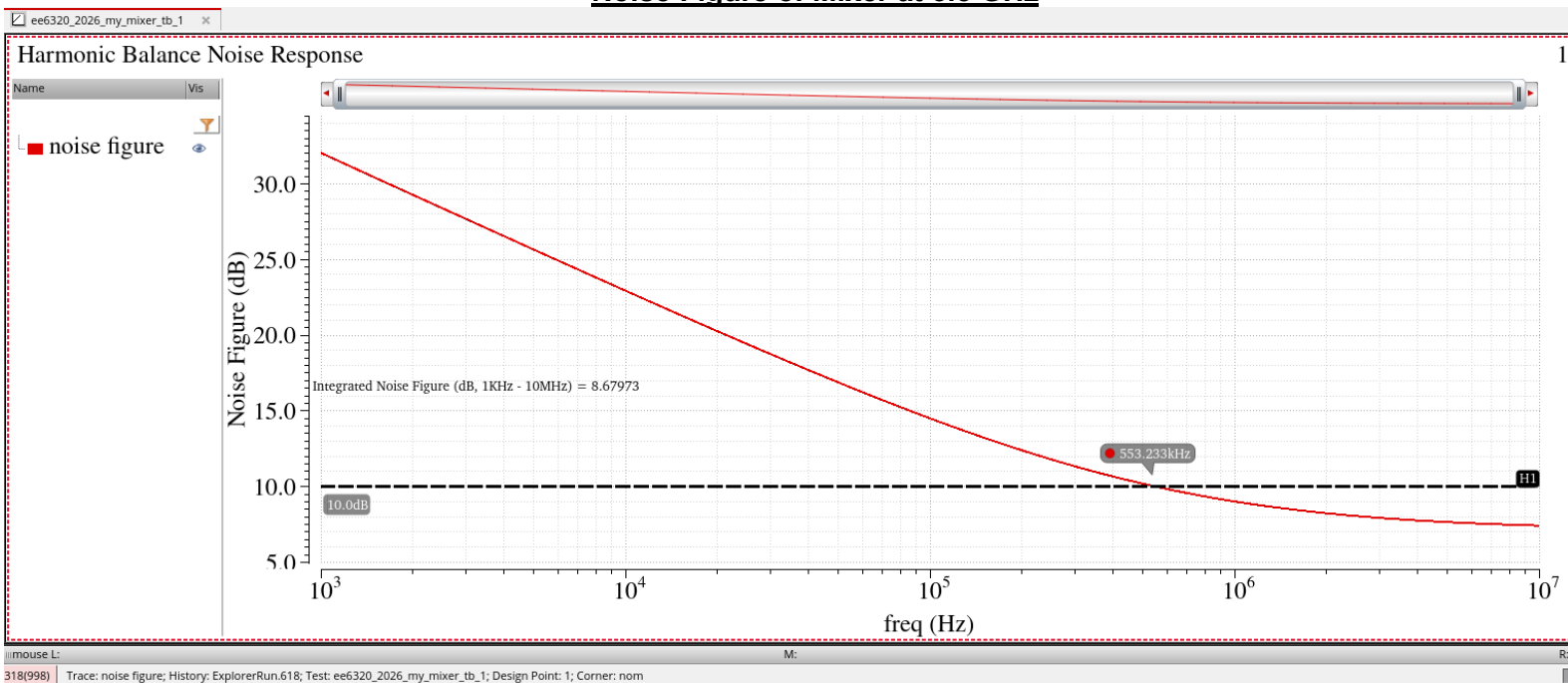
Fixed Constant Parameters

- Transconductance MOS C_{gs} : 96.79 fF
- Current Mirror MOS: $W = 0.7 \mu\text{m}$, $L = 0.12 \mu\text{m}$
- Coupling capacitances: 1 μF (each)
- Bias Resistances: 100 k Ω (each)

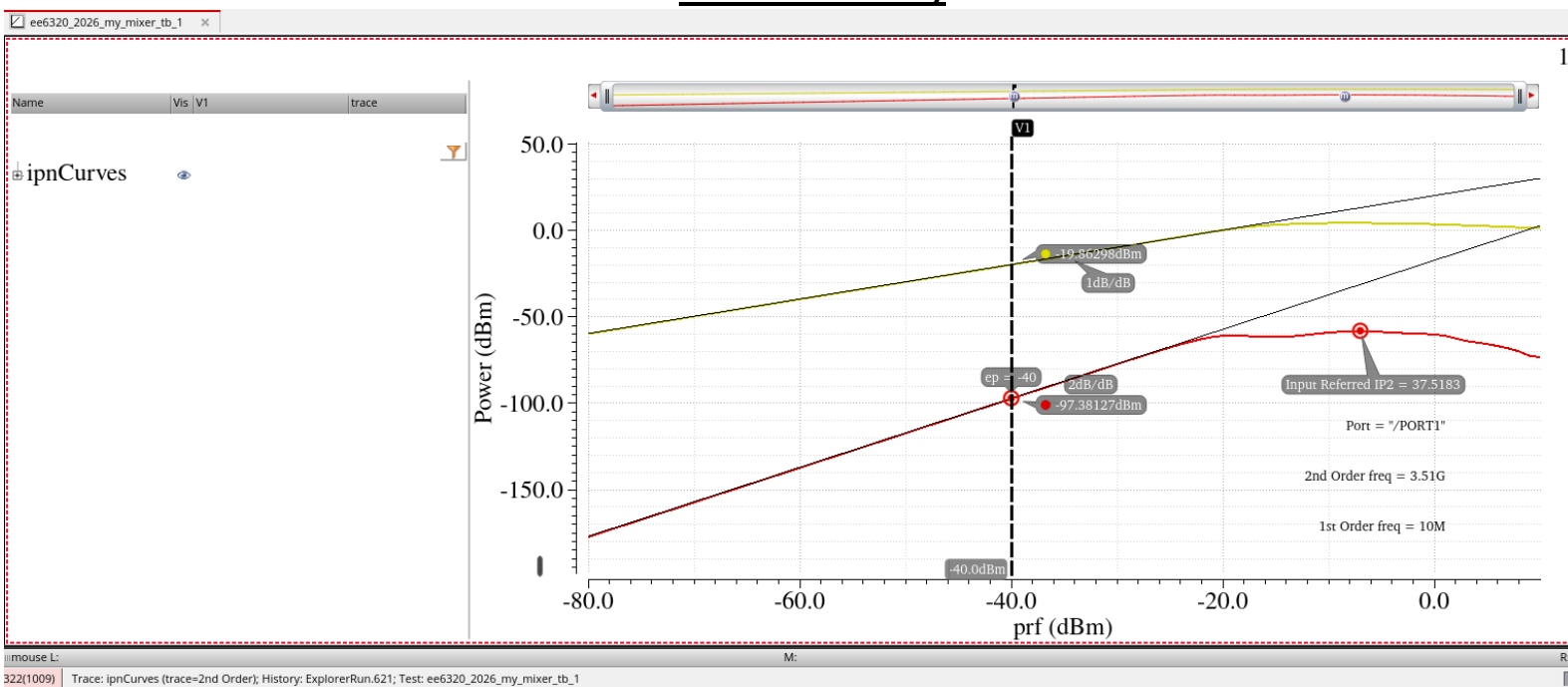
Conversion Gain of Mixer at 3.5 GHz



Noise Figure of Mixer at 3.5 GHz

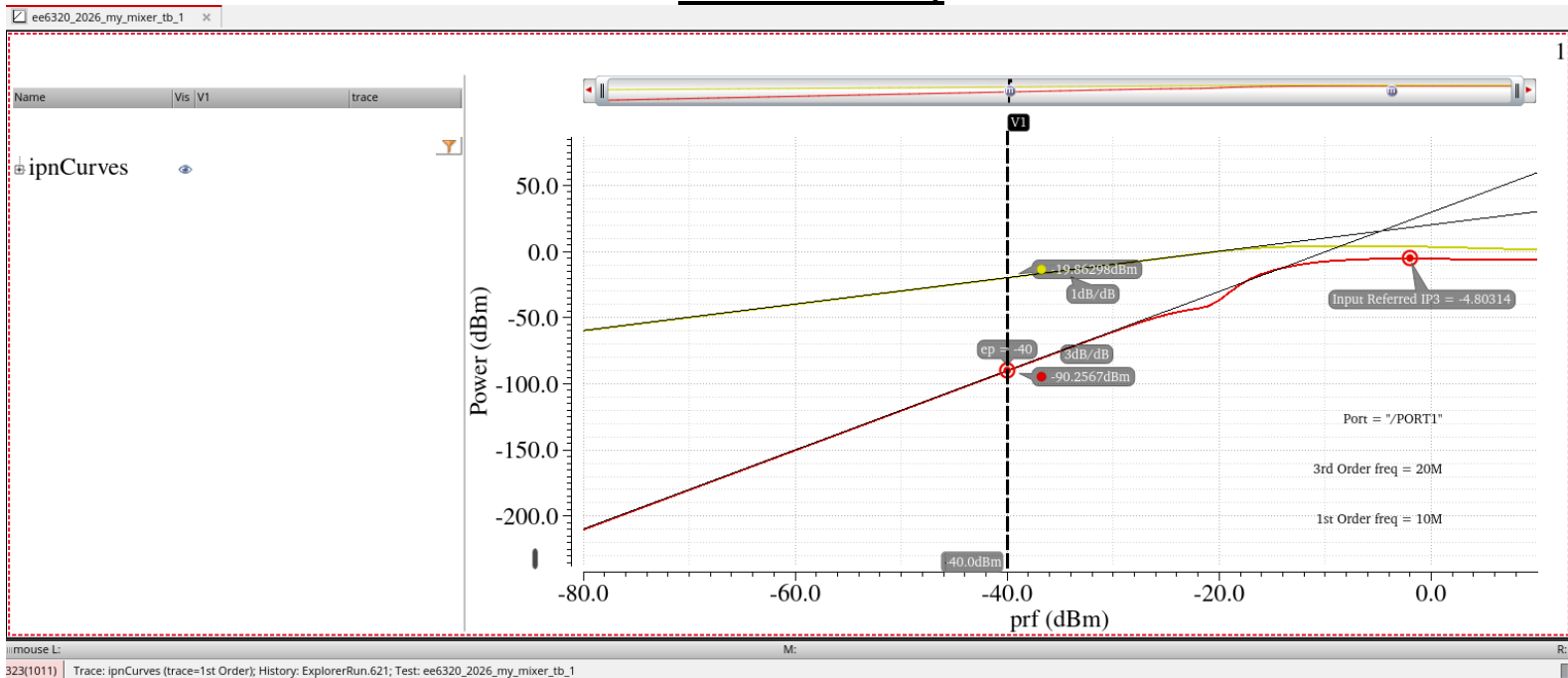


Mixer IIP2 Linearity



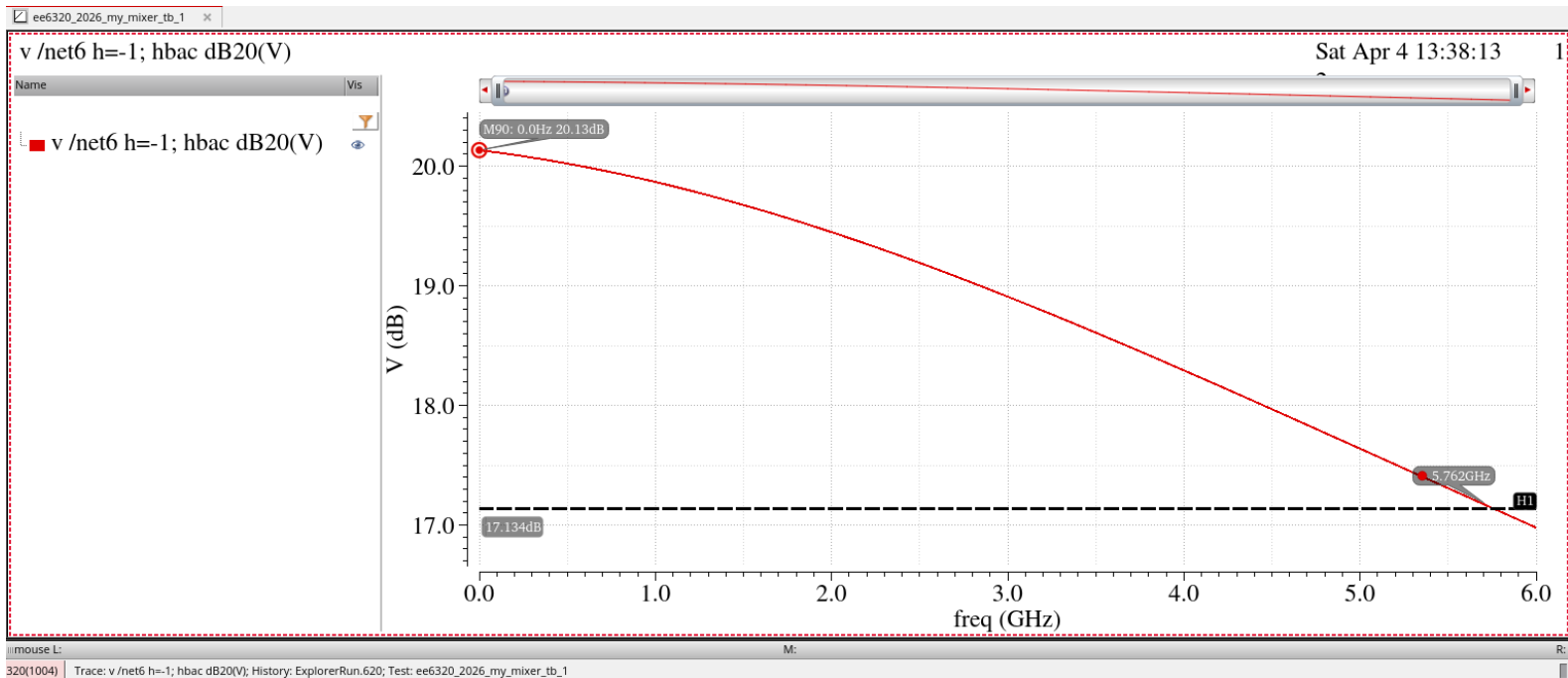
Fundamental tone (10M): $\text{frf}_2 - \text{flo}$; IM₂ tone (3.51G): $\text{frf}_2 + \text{frf}_1 - \text{flo}$

Mixer IIP3 Linearity

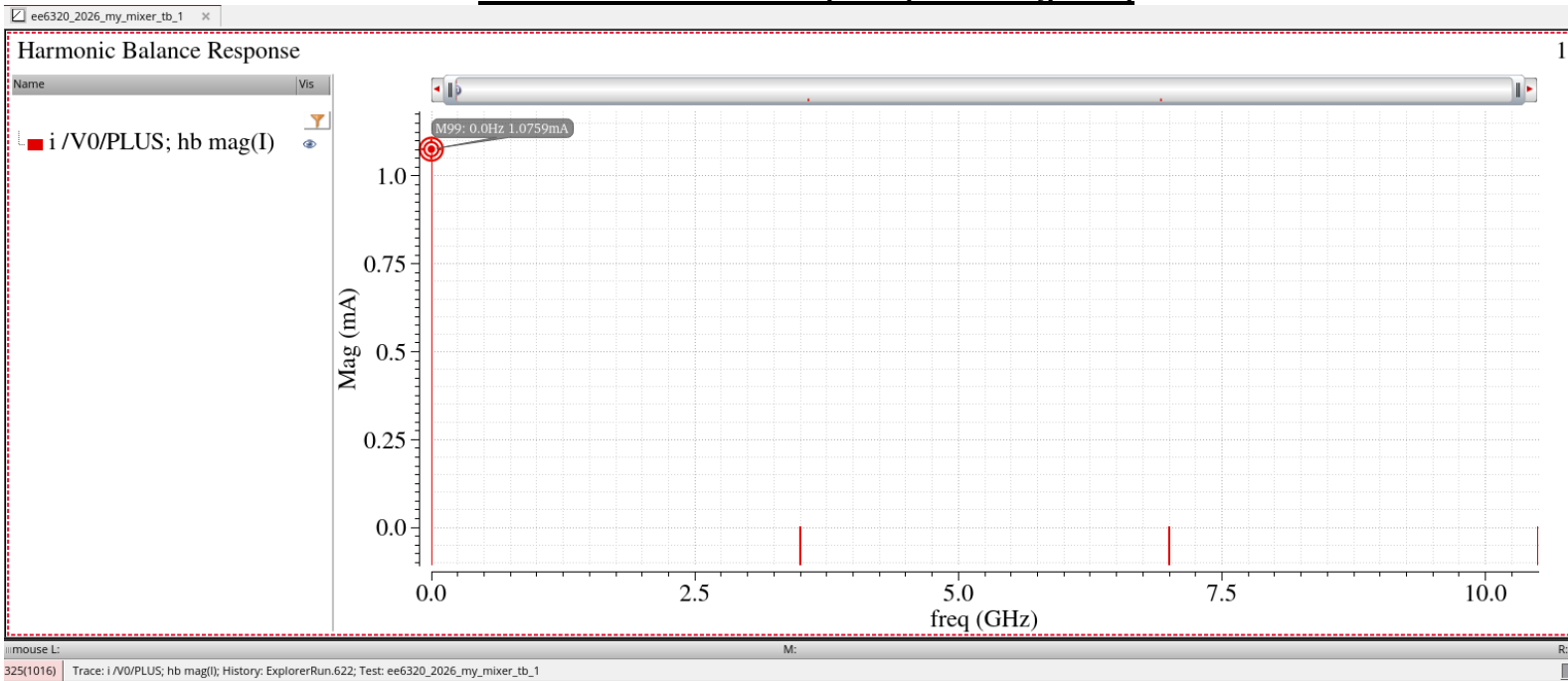


Fundamental tone (10M): $\text{frf}_2 - \text{flo}$; IM_3 tone (20M): $2 * \text{frf}_2 - \text{frf}_1 - \text{flo}$

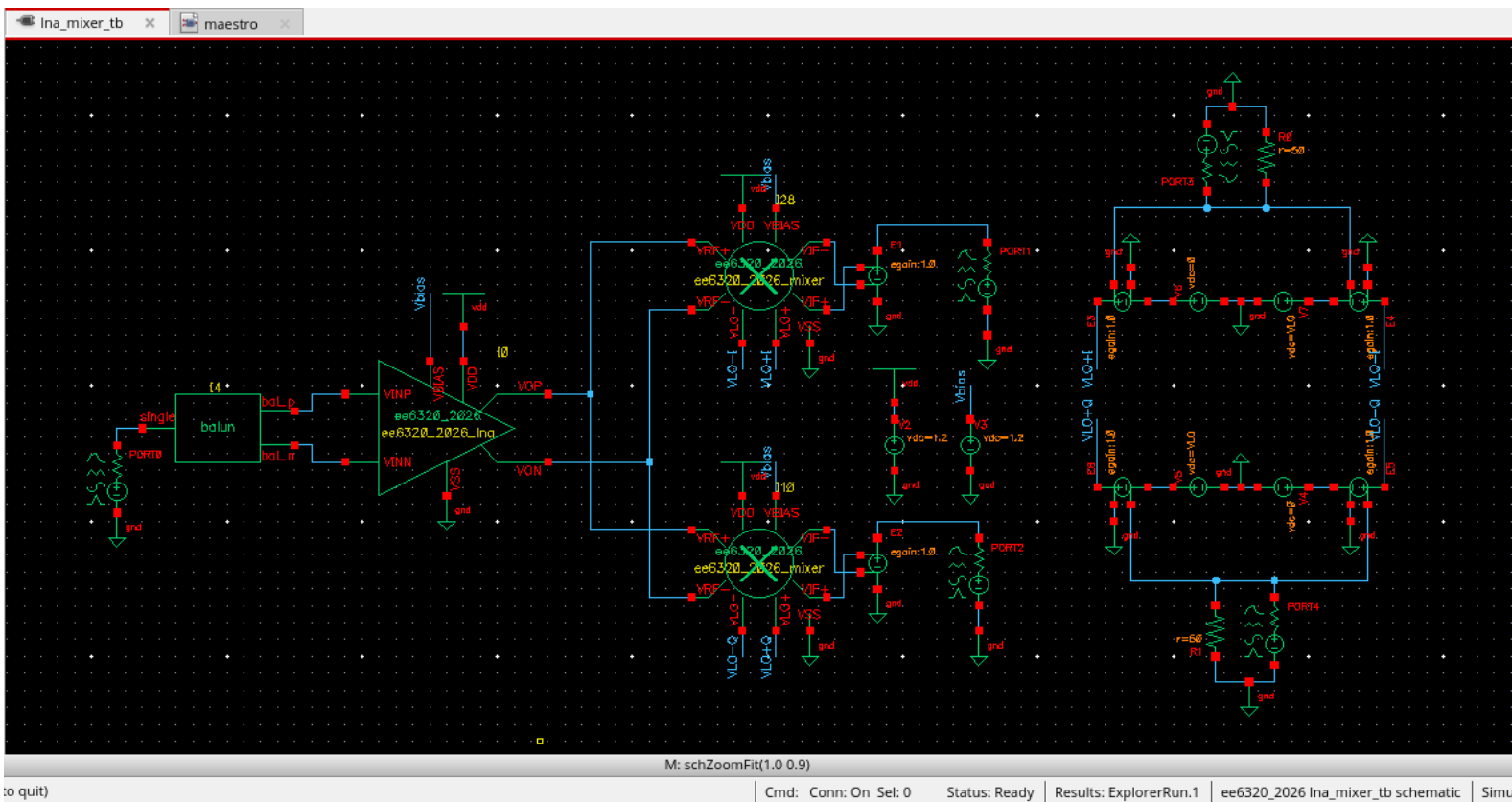
Mixer 3-dB Bandwidth



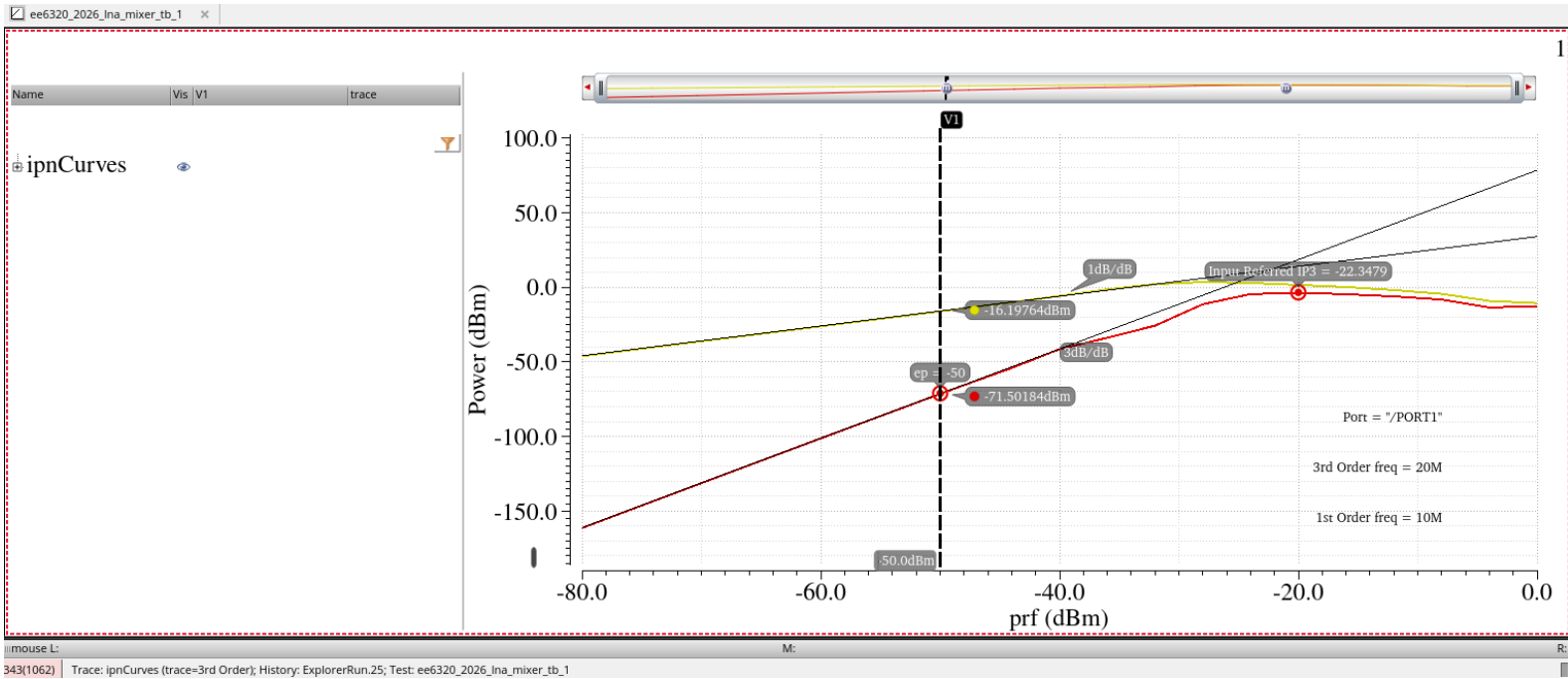
Mixer DC Power Consumption [Excluding Bias]



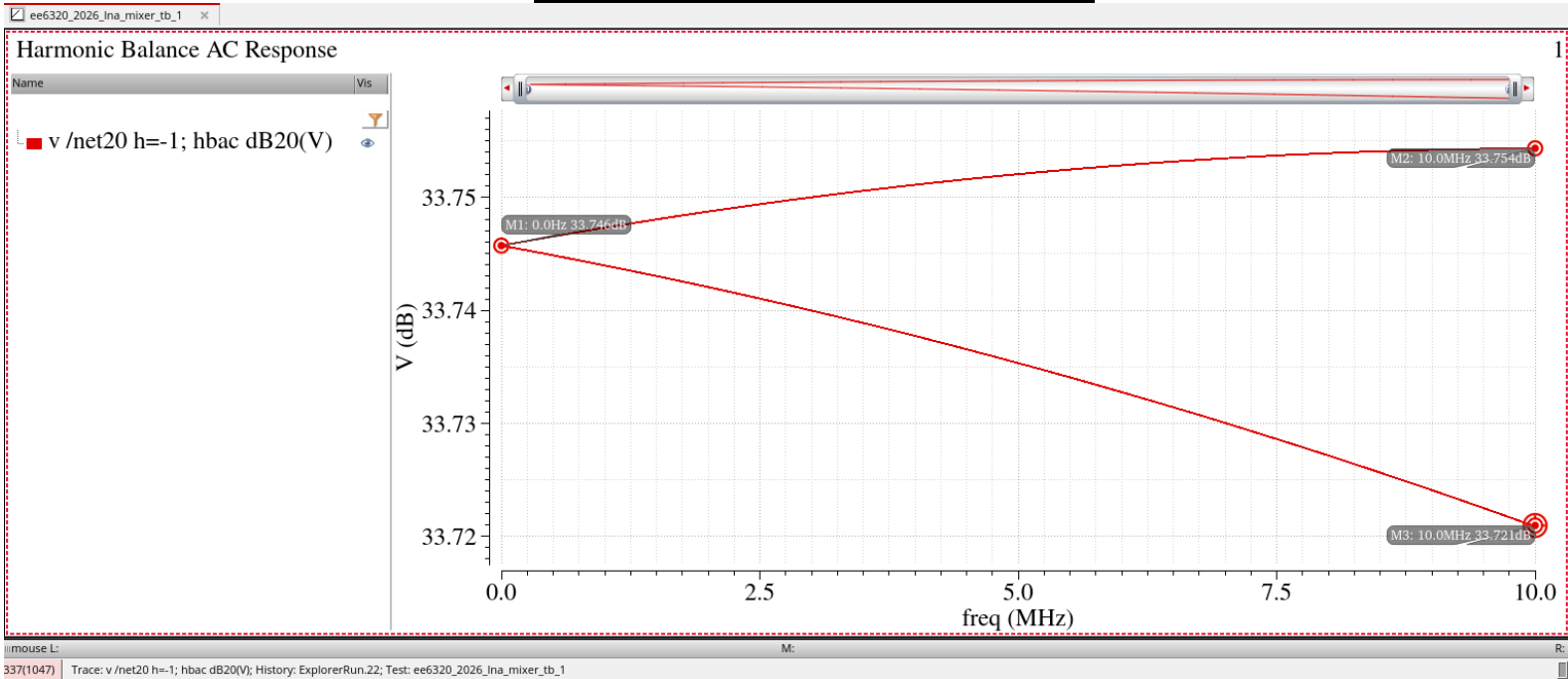
LNA + Mixer Testbench



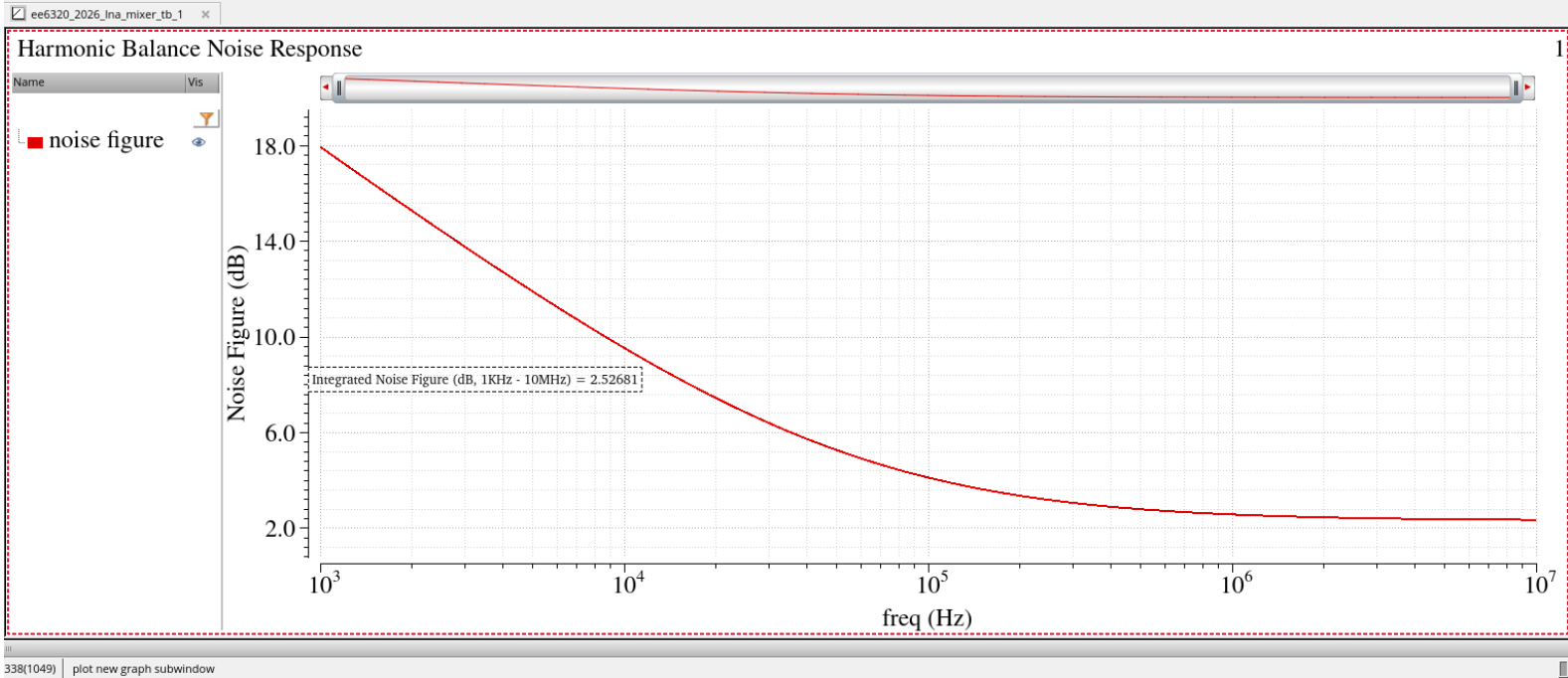
LNA + Mixer IIP3 Linearity



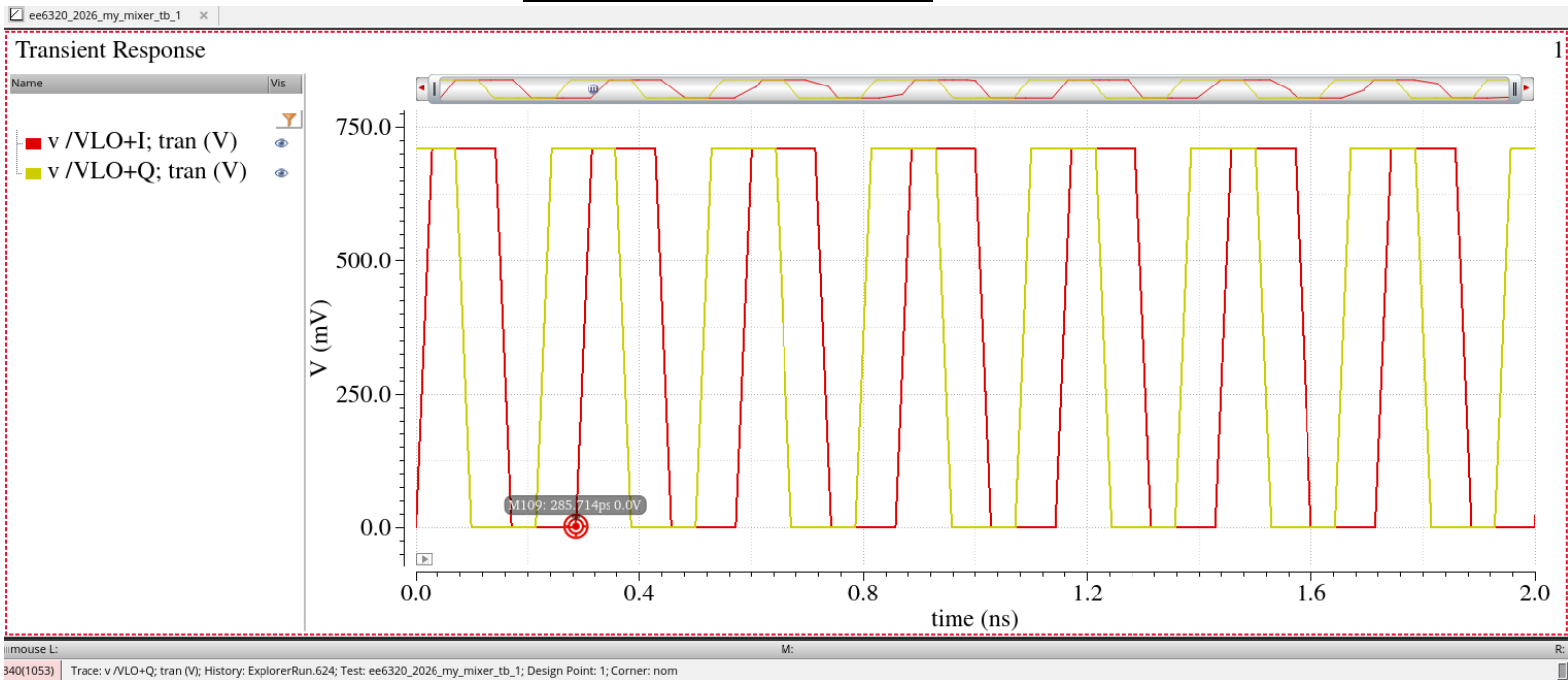
LNA + Mixer Conversion Gain at 3.5 GHz



LNA + Mixer Noise Figure at 3.5 GHz



LO Waveform Characteristics



Characteristics taken are (for generalized manner, as f_{lo} varies for various analysis):

- Frequency: f_{lo} Hz
- Time period: $1/f_{lo}$ s
- Delay time: 0 (for I Lo signal), $0.25/f_{lo}$ s (for Q Lo signal)
- Pulse Width: $0.4/f_{lo}$ s
- Rise/Fall Time: $0.1/f_{lo}$ s

Calculations done for the Project

1) gm_1

- For IBM 0.13μm, $\epsilon_{ox} = 3.45 \times 10^{-11}$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11}}{32 \times 10^{-10}} = 0.0108 \text{ F/m}^2$$

- $\mu_n = 0.033 \text{ m}^2/\text{Vs}$ (from simulation)
- $I_1 = 0.39 \text{ mA}$

$$\frac{I_1}{I_b} = \frac{W_1}{W_0} = 100 \quad (\text{where } I_b: \text{bias current, } M_0: \text{current mirror MOSFET})$$

- $g_{m1} = 12.73 \text{ mS}$

$$g_{m1} = \sqrt{2 \mu_n C_{ox} \frac{W_1}{L_1} I_1}$$

- But, from the actual simulation, $g_{m1} = 5.898 \text{ mS}$

2) Gain of mixer (G_2)

- $G_2 = 15.49 = 23.8 \text{ dB}$

$$G_2 = \frac{2}{\pi} g_{m1} R_L \quad (R_L = 1120 \Omega)$$

- But, from simulation ($g_{m1} = 5.989$), $G_2 = 4.27 = 12.6 \text{ dB}$

3) Gain of the cascade (G)

$$G = G_1 \times G_2$$

$$G \text{ (dB)} = G_1 \text{ (dB)} + G_2 \text{ (dB)}$$

4) NF of cascade (NF)

- $F_1 = 1.806 \text{ dB} = 1.515$
- $G_1 = 21.818 \text{ dB} = 12.328$
- $F_2 = 8.68 \text{ dB} = 7.379$
- From Friis equation

$$F = F_1 + \frac{F_2 - 1}{G_1} = 1.515 + \frac{6.379}{12.328} = 2.032$$

- $\text{NF} = 10 \log_{10}(F) = 3.08 \text{ dB}$ expected

5) IIP3 of cascode

- $\text{IIP}_{3,1} = -9.77 \text{ dB} = 0.1054$

$$A^2 \text{IP}_{3,1} = \text{IIP}_{3,1} \times 2 R_s = 10.54$$

- $\text{IIP}_{3,2} = -4.80 \text{ dB} = 0.3311$

$$A^2 \text{IP}_{3,2} = \text{IIP}_{3,2} \times 2 R_s = 33.11$$

- $G_1 = 21.818\text{dB} = 12.328$
- From Friis equation

$$\frac{1}{A^2IP_3} = \frac{1}{A^2IP_{3,1}} + \frac{G_1^2}{A^2IP_{3,2}} = \frac{1}{10.54} + \frac{12.328^2}{33.11} = 4.68$$

$$A^2IP_3 = 0.213$$

$$IIP_3 = \frac{A^2IP_3}{2 R_S} = 0.00213 = -26.7 \text{ dB expected}$$

Procedure adopted for the Project

- After plugging in the values obtained from hand calculations, we simulate the gain and NF.
- If the NF doesn't meet the specs, we can increase the gm of M1 by either increasing its current or by increasing its width.
- If the gain doesn't meet the specs, we can increase the load resistance RL.
- When both the NF and gain specs are met, we plot the linearity, that is IIP2 and IIP3.
- If they are not met, making the triode limits of M1 better improves the IIP3. This can be done by –
 - Increasing VLO (worsens gain expansion, may worsen the overall IIP3 depending on which MOSFET is the major contributor)
 - Increasing w of M1 (worsens gain expansion)
 - Increasing the w of the switches M2,3.
 - Decreasing the current of M1 (reduces gain and NF)
- If the gain expansion is a problem, it can be reduced by improving the triode limits of the switches M2,3 by the following methods –
 - Reducing RL (worsens gain)
 - Reducing VLO (may worsen the overall IIP3 depending on which MOSFET is the major contributor)
 - Reducing the bias current, that is the current through M1 (worsens gain and NF)
 - Reducing w of M1 (worsens gain, NF and IIP3)
- The specifications that have been worsened by attempting to improve another specification need to be compensated for, till an optimal balance is reached between the involved specs.
- After all specifications are met, power consumption can be optimized by reducing the bias current just enough such that all specs are barely met.
- Other circuit techniques can be explored, such as the following –
 - Source degeneration (differential) – Improves linearity, but can only be done if the available gain is much higher than the spec (because gain changes from $\frac{2}{\pi} gm_1 RL$ to $\frac{2}{\pi} \frac{RL}{Re}$, where $gm_1 > \frac{1}{Re}$).
 - PMOS loads with CMFB using resistors – Improves the gain, but linearity will be a problem.
 - PMOS loads with CMFB using opamp – Improves the gain and linearity, but system will be unstable and have to be compensated with an appropriate capacitor.
 - Bleeder current sources – Can either improve or worsen gain and linearity, depending on the point of operation
 - Multi tanh transconductor – Improves linearity, but worsens power consumption

Path to the Project Files

Mixer schematic - /home/ep22b020/cadence/EE6320_2026/ee6320_2026/ee6320_2026_mixer